

Merge, Append and Intersect

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Data

All the data is the shared Google Drive under Merge, Append and Intersect and include. . .

4 shapefiles

- NOAA_Habitat_Map
- VI_National_Monument
- VI_National_Park
- sttstj_sampleframe

1 .csv file

- sttstj_benthic_sites_2021.csv

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The exercises today will use the new tools we discussed in lecture as well as previous tools you already learned.

Part 1

1. Import all the data into your project directory (4 shapefiles).
 - What is the area (km^2) of Coral Reef and Colonized Hardbottom in each of the unique Marine Protected Areas (national parks and monument)? Hint: Inspect each shapefile separately, what are the attributes of each, could they be renamed for clarity, think through the entire workflow first before you start geoprocessing, write it out if needed, the order of operations will be important

Part 2

1. Create a shapefile (point) from the sttstj_benthic_sites_2021.csv dataset.
2. The sttstj_sampleframe is a 50x50m gridded data set that serves as NCRMP's sampling domain. It contains additional metadata that the actual sampling points do not have. The sampling frame further divides St Thomas/John into SUB_REGIONS and administrative zones (ADMIN) to help allocate our sampling effort.
 - How many benthic sites (PRIMARY_SAMPLE_UNITS) were sampled in 2021 in each ADMIN? How many benthic sites were sampled in each SUB_REGION?